Analytical Report

The primary objective of this housing.csv dataset is to predict the housing price based on certain factors like house area, bedrooms, furnished, nearness to main road, etc. Variable description is provided below:

area = area of a house
bedrooms = number of house bedrooms
bathrooms = number of house bathrooms
stories = number of house stories
mainroad = whether connected to the main road
guestroom = whether has a guestroom
basement = whether has a basement
hotwaterheating = whether has hot water heater
airconditioning = whether has an airconditioning
parking = the number of parking
price = price of the house

Step 1: Importing data and converting sting variables into dummy variables.

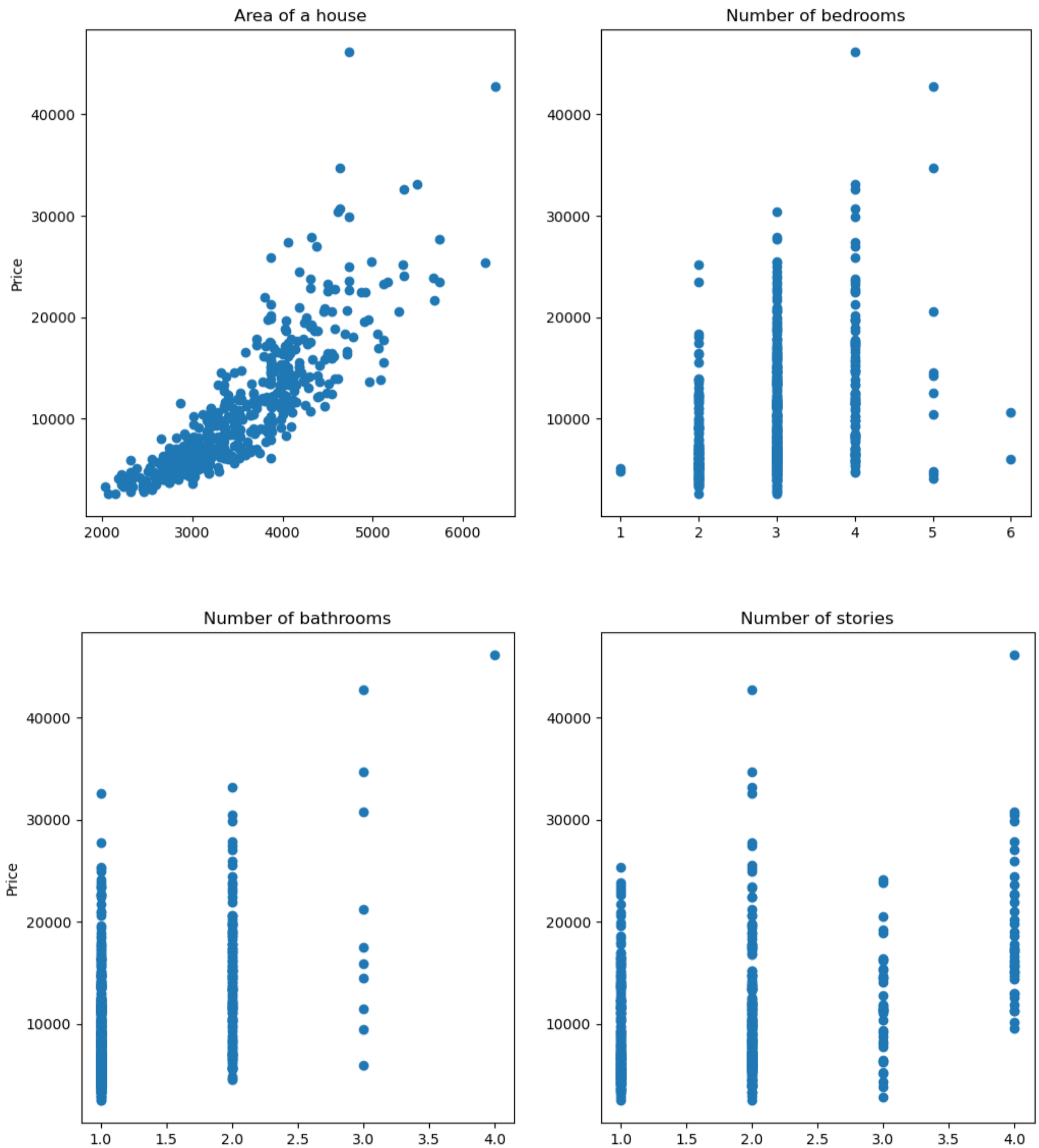
This is the original data:

	area	bedrooms	bathrooms	stories	mainroad	guestroom	basement	hotwaterheating	airconditioning	parking	price
0	4306.971093	4	2	3	yes	no	no	no	yes	2	23807.62253
1	4732.863826	4	4	4	yes	no	no	no	yes	3	46173.57355
2	4989.989980	3	2	2	yes	no	yes	no	no	2	25491.22063
3	4330.127019	4	2	2	yes	no	yes	no	yes	3	18941.80522
4	4306.971093	4	1	2	yes	yes	yes	no	yes	2	17582.99123

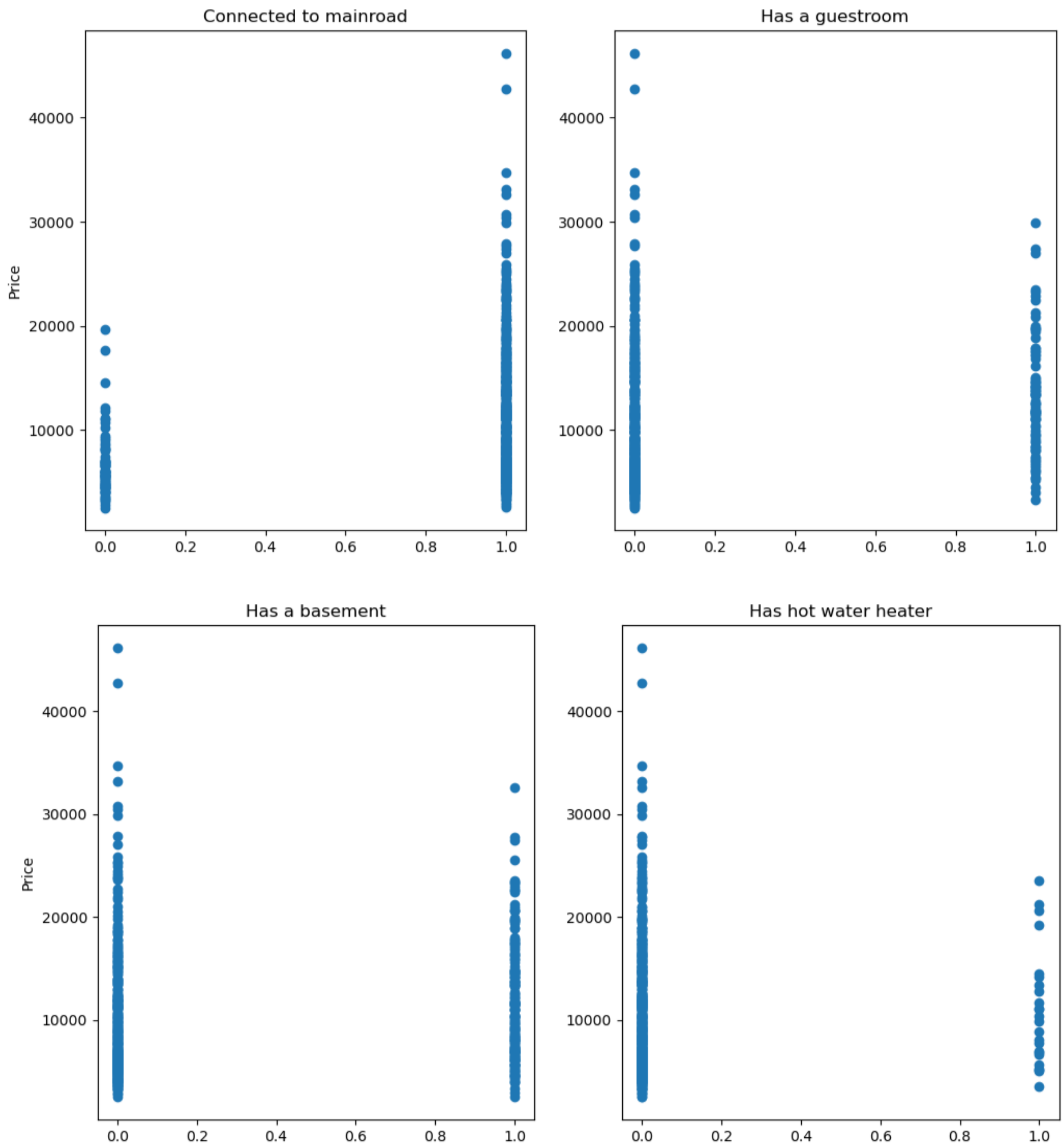
After converting into dummy variables:

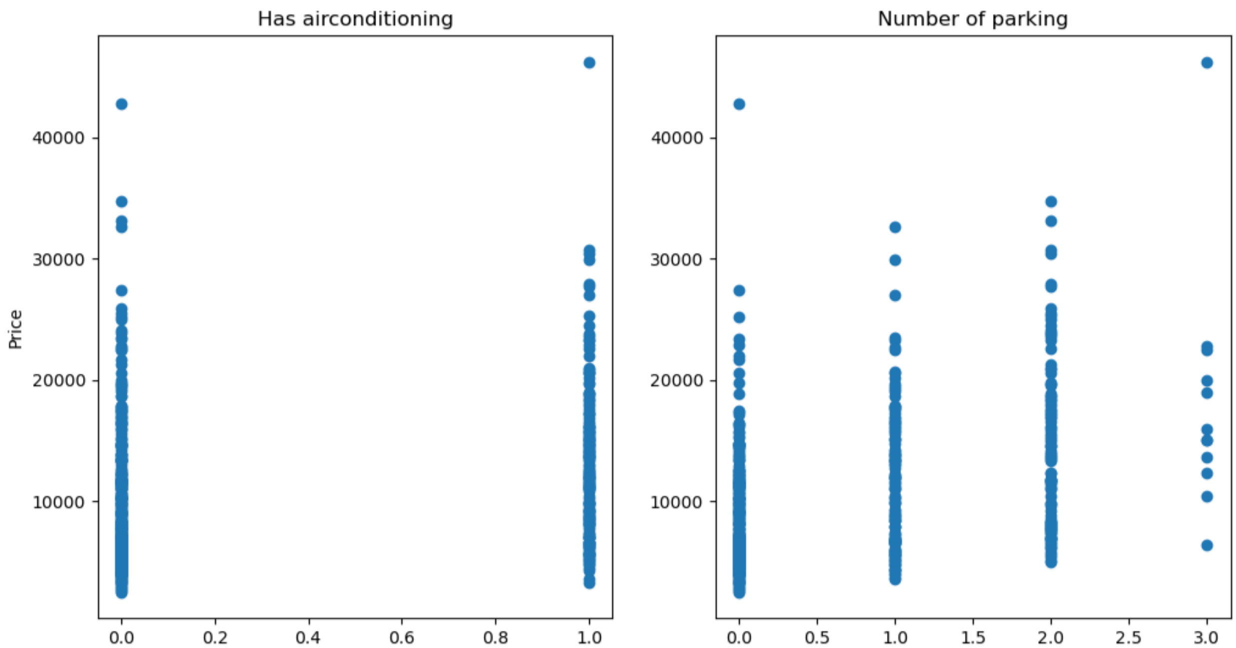
	area	bedrooms	bathrooms	stories	mainroad	guestroom	basement	hotwaterheating	airconditioning	parking	price
0	4306.971093	4	2	3	1	0	0	0	1	2	23807.62253
1	4732.863826	4	4	4	1	0	0	0	1	3	46173.57355
2	4989.989980	3	2	2	1	0	1	0	0	2	25491.22063
3	4330.127019	4	2	2	1	0	1	0	1	3	18941.80522
4	4306.971093	4	1	2	1	1	1	0	1	2	17582.99123

Step 2: We construct scatter plots between dependent variable with each of independent variable



Area of a house
Number of bedrooms
Number of bathrooms
Number of stories
Price
Price
10000
20000
30000
40000
2000
3000
4000
5000
6000
1
2
3
4
5
6
1.0
1.5
2.0
2.5
3.0
3.5
4.0


Connected to mainroad
Has a guestroom
Has a basement
Has hot water heater
Price
Price
40000
30000
20000
10000
0.0
0.2
0.4
0.6
0.8
1.0



The scatter plots for all independent variables, except the area of the house, suggest that they are discrete variables and show a generally positive relationship with housing price. However, there is some variation in housing prices within each category of these variables. The area of the house shows a slightly positive curved line, which is acceptable and may be considered to be a linear regression relationship with housing price.

Step 3: Fit the values into regression line and construct OLS Regression Result tables. Construct diagnostic checks for the dataset

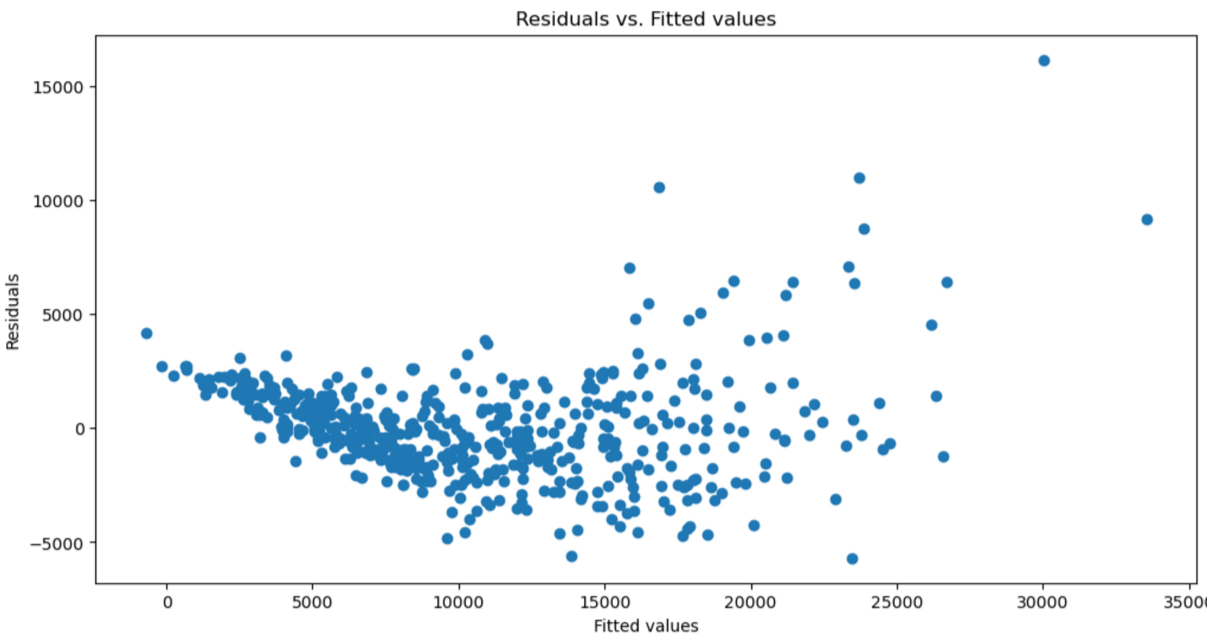
```
OLS Regression Results
==============================================================================
Dep. Variable:                  price   R-squared:                       0.868
Model:                            OLS   Adj. R-squared:                  0.865
Method:                 Least Squares   F-statistic:                     346.4
Date:                Thu, 05 Mar 2026   Prob (F-statistic):         7.67e-225
Time:                        20:35:14   Log-Likelihood:                -4935.6
No. Observations:                 540   AIC:                             9893.
Df Residuals:                     529   BIC:                             9940.
Df Model:                          10
Covariance Type:            nonrobust
===================================================================================
                      coef    std err          t      P>|t|      [0.025      0.975]
-----------------------------------------------------------------------------------
const            -1.99e+04    631.814    -31.500      0.000   -2.11e+04   -1.87e+04
area                6.5789      0.158     41.578      0.000       6.268       6.890
bedrooms          308.4766    154.663      1.995      0.047       4.647     612.306
bathrooms        2400.1873    221.901     10.816      0.000    1964.273    2836.102
stories          1702.2485    137.413     12.388      0.000    1432.306    1972.191
mainroad         -551.6684    301.545     -1.829      0.068   -1144.041      40.704
guestroom        -188.2469    284.738     -0.661      0.509    -747.603     371.109
basement         1331.7594    232.131      5.737      0.000     875.747    1787.772
hotwaterheating  -358.8964    483.557     -0.742      0.458   -1308.824     591.031
```

```
airconditioning  -627.6635    232.231     -2.703      0.007   -1083.871    -171.456
parking           775.0578    125.123      6.194      0.000     529.258    1020.857
==============================================================================
Omnibus:                      197.678   Durbin-Watson:                   1.748
Prob(Omnibus):                  0.000   Jarque-Bera (JB):             1321.689
Skew:                           1.444   Prob(JB):                    9.97e-288
Kurtosis:                      10.099   Cond. No.                     2.35e+04
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly spec
ified.
[2] The condition number is large, 2.35e+04. This might indicate that there are
strong multicollinearity or other numerical problems.
```

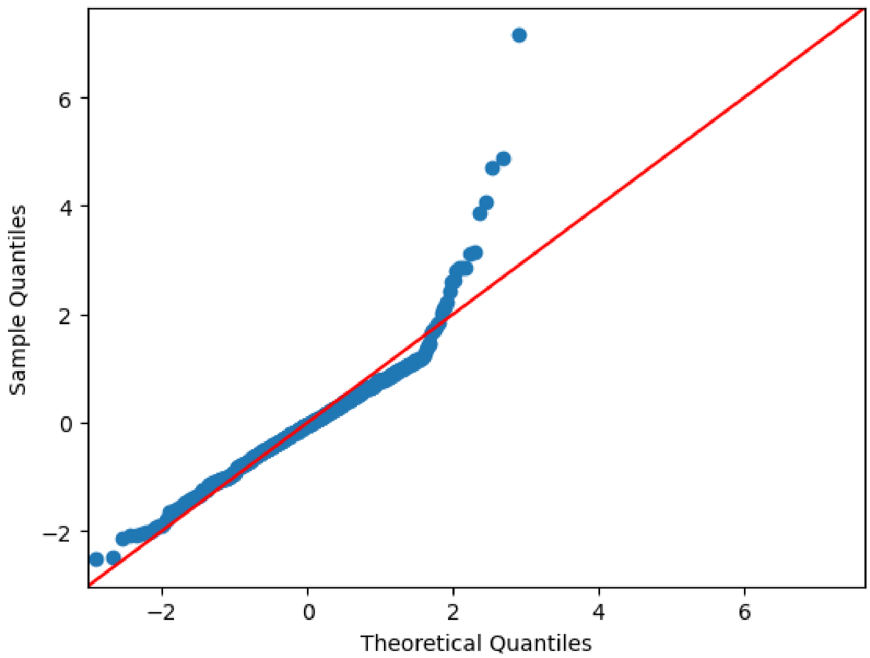
According to the result table, mainroad variable is marginally significant from the 0.05 significant value. Guestroom and hotwaterheating are not significant variables that can affect to our dependent variable. Other than those mentioned, the remaining independent variables significantly affect the price.

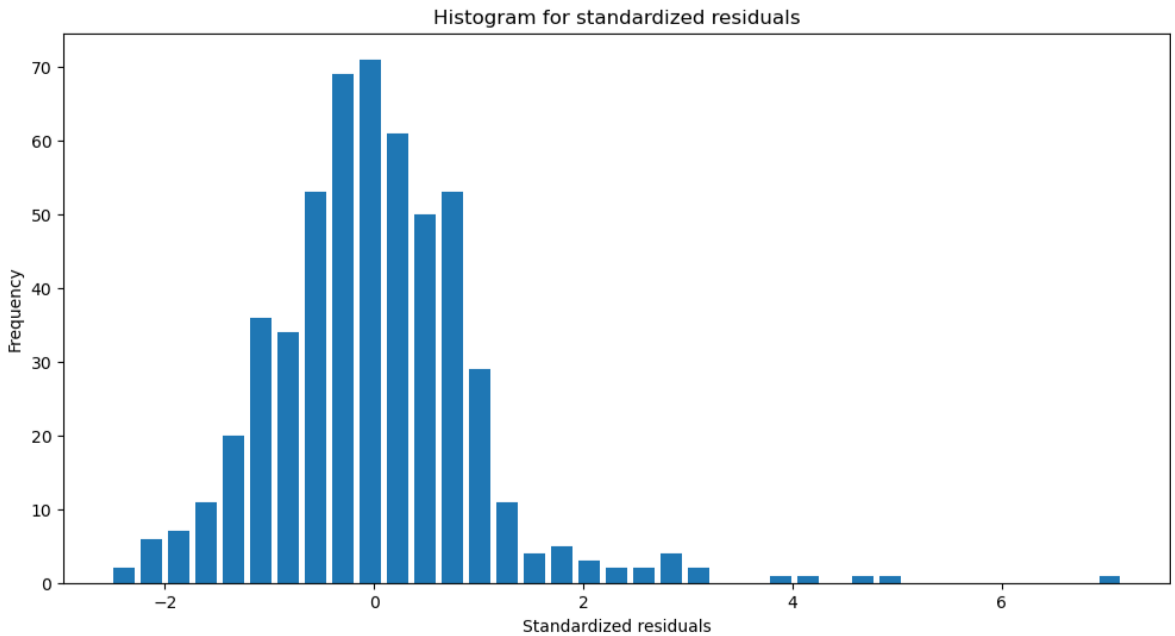
The regression line:
price = -1.99e+04 + 6.5789area + 308.4766bedrooms + 2400.1873bathrooms +1702.2485stories – 551.6684mainroad – 188.2469guestroom + 1331.7594basement – 358.8964hotwaterheating – 627.6635airconditioning

R-squared: 0.868, indicating a good explanation of the variables by regression line
AIC: 9893
BIC: 9940



The residual plot shows clear violations in zero mean and constant variance consumptions. Moreover, it suggests there are potential outliers on the right tail of the plot.





The QQ plot shows that the residuals deviate from the red reference line, particularly in the right tail, suggesting a normality violation and indicating right skewness. The pattern is also represented in histogram plot, where we can see the right skewness caused by potential outliers, which observations lie more than 3 standard deviations away from the mean.

The residual plots from initial regression indicate heteroskedasticity and right-skewed residuals. To stabilize variance and improve the normality of residuals, a log transformation of the dependent variable (price) was applied.

<u>Step 4: Apply log transformation for dependent variable. Construct fitted regression and diagnostic checks for the new dataset</u>

This is the data after log transformation for dependent variable:

	area	bedrooms	bathrooms	stories	mainroad	guestroom	basement	hotwaterheating	airconditioning	parking	log_price
0	4306.971093	4	2	3	1	0	0	0	1	2	10.077761
1	4732.863826	4	4	4	1	0	0	0	1	3	10.740163
2	4989.989980	3	2	2	1	0	1	0	0	2	10.146089
3	4330.127019	4	2	2	1	0	1	0	1	3	9.849127
4	4306.971093	4	1	2	1	1	1	0	1	2	9.774687

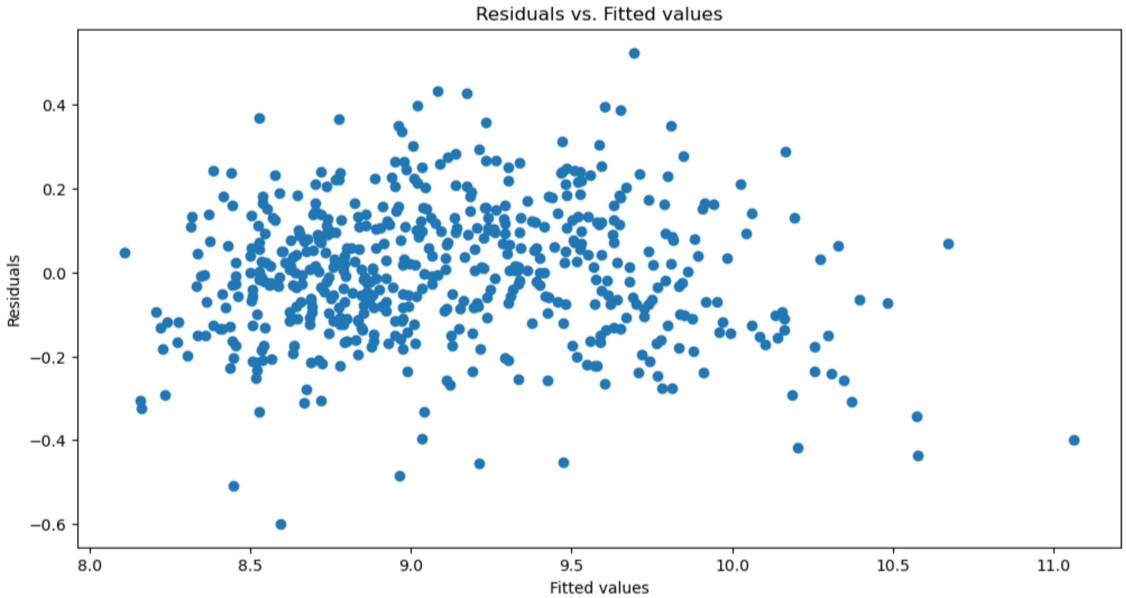
```
OLS Regression Results
==============================================================================
Dep. Variable:              log_price   R-squared:                       0.910
Model:                            OLS   Adj. R-squared:                  0.909
Method:                 Least Squares   F-statistic:                     537.2
Date:                Thu, 05 Mar 2026   Prob (F-statistic):          1.26e-269
Time:                        20:35:29   Log-Likelihood:                 218.62
No. Observations:                 540   AIC:                            -415.2
Df Residuals:                     529   BIC:                            -368.0
Df Model:                          10
Covariance Type:            nonrobust
===================================================================================
                      coef    std err          t      P>|t|      [0.025      0.975]
-----------------------------------------------------------------------------------
const               6.3858      0.045    141.216      0.000       6.297       6.475
area                0.0006   1.13e-05     51.754      0.000       0.001       0.001
bedrooms            0.0373      0.011      3.372      0.001       0.016       0.059
bathrooms           0.1541      0.016      9.704      0.000       0.123       0.185
stories             0.1405      0.010     14.283      0.000       0.121       0.160
mainroad            0.0160      0.022      0.741      0.459      -0.026       0.058
guestroom           0.0250      0.020      1.229      0.220      -0.015       0.065
basement            0.1497      0.017      9.013      0.000       0.117       0.182
hotwaterheating     0.0243      0.035      0.702      0.483      -0.044       0.092
airconditioning    -0.0123      0.017     -0.738      0.461      -0.045       0.020
parking             0.0600      0.009      6.695      0.000       0.042       0.078
==============================================================================
Omnibus:                        4.442   Durbin-Watson:                   1.955
Prob(Omnibus):                  0.108   Jarque-Bera (JB):                4.840
Skew:                          -0.118   Prob(JB):                       0.0889
Kurtosis:                       3.400   Cond. No.                     2.35e+04
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly spec
ified.
[2] The condition number is large, 2.35e+04. This might indicate that there are
strong multicollinearity or other numerical problems.
```

After the log transformation of the dependent variable, the result shows that mainroad, guestroom, hotwaterheating and airconditioning are not significant variables. However, the R-squared is improved slightly significant from 0.868 to 0.910.

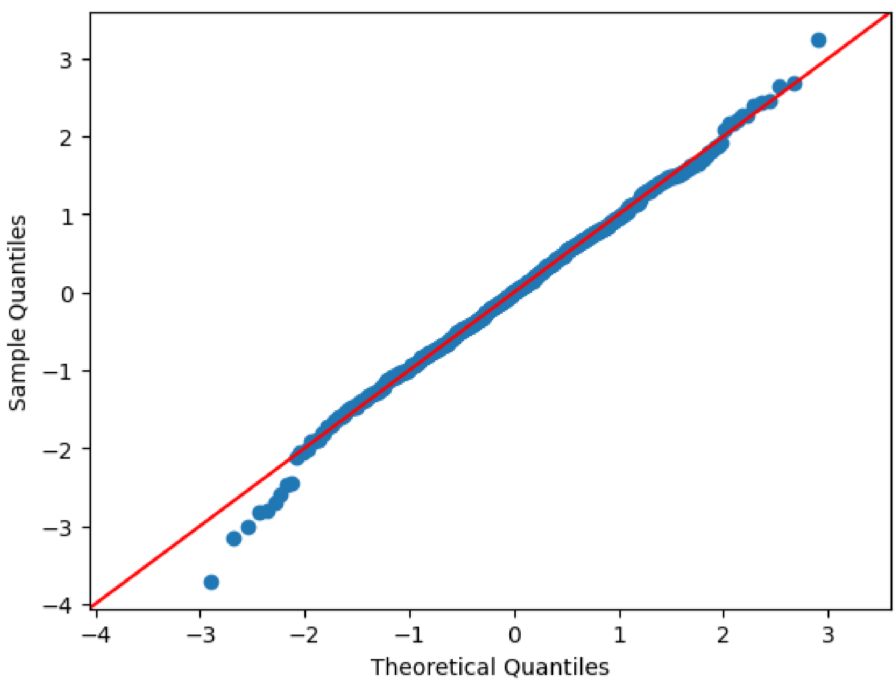
The regression equation:

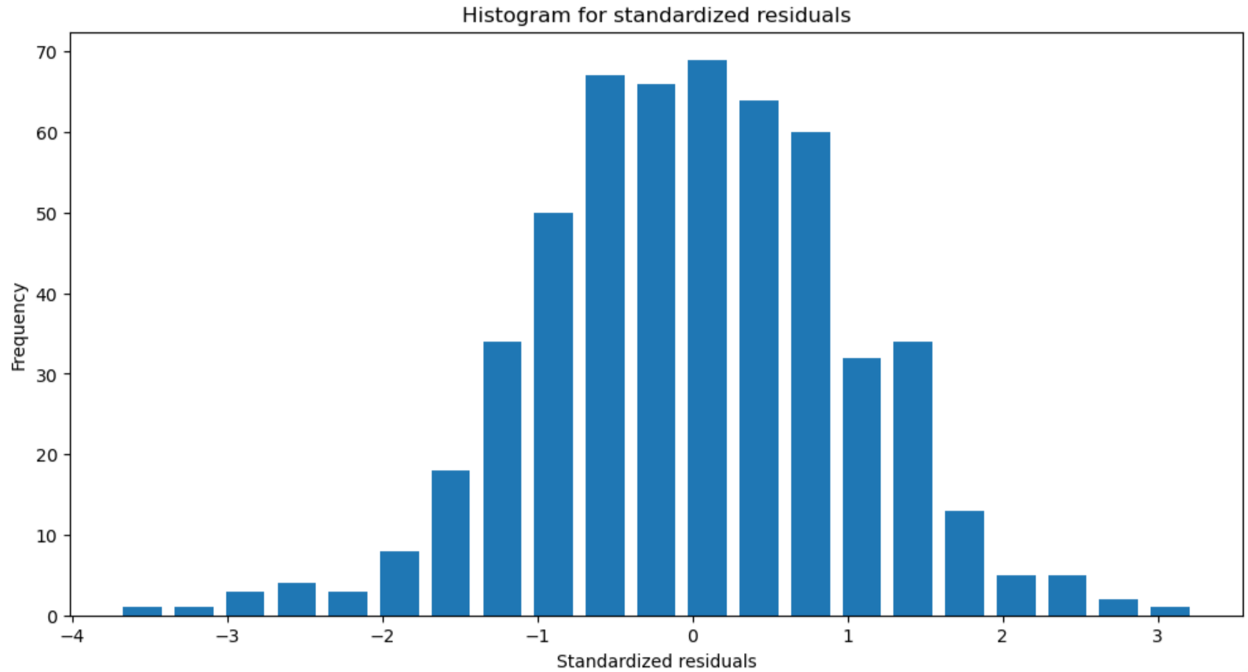
log_price = 6.3858 + 0.0006area + 0.0373 bedrooms + 0.1541bathrooms + 0.1405stories + 0.0160mainroad + 0.025guestroom + 0.1497basement + 0.0243hotwaterheating – 0.0123airconditioning + 0.06parking

R-squared: 0.910
AIC = -415.2
BIC = -368



The residuals plot suggests that the log transformation has improved the stability of the variance, as the residuals deviate randomly and have constant distance from 0. indicating there are no violation of zero mean or constant variance assumptions of the regressions model.





The QQ plot and histogram plot both show slight violation of normality after log transformation and some potential residuals deviate on the left around 4 standard deviation away from the mean, indicating there is still marginally left skewness.
From the OLS result, the large condition number suggests potential multicollinearity among explanatory variables. To construct a more accruate model and retain only significant predictors, forward feature selection is applied.

<u>Step 5: Construct forward feature selection and display the result</u>

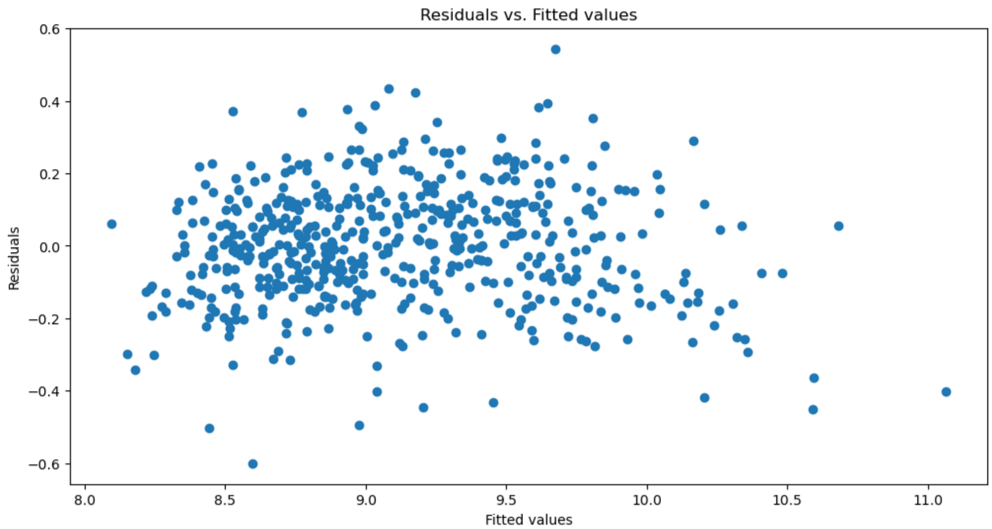
```
                                                model        RSS  \
1   <statsmodels.regression.linear_model.Regressio...  35.086933
2   <statsmodels.regression.linear_model.Regressio...  23.920680
3   <statsmodels.regression.linear_model.Regressio...  18.937242
4   <statsmodels.regression.linear_model.Regressio...  15.758463
5   <statsmodels.regression.linear_model.Regressio...  14.445373
6   <statsmodels.regression.linear_model.Regressio...  14.154952
7   <statsmodels.regression.linear_model.Regressio...  14.115968
8   <statsmodels.regression.linear_model.Regressio...  14.096428
9   <statsmodels.regression.linear_model.Regressio...  14.082198
10  <statsmodels.regression.linear_model.Regressio...  14.069106

    Rsquared_adj         AIC         BIC
1            NaN   60.233821   68.816959
2            NaN -144.632304 -131.757597
3            NaN -268.783283 -251.617007
4            NaN -366.009830 -344.551985
5            NaN -410.991690 -385.242275
6            NaN -419.958870 -389.917886
7            NaN -419.448129 -385.115576
8            NaN -418.196150 -379.572028
9            NaN -416.741541 -373.825850
10           NaN -415.243800 -368.036539
Foward selection take:  0.4360806941986084 seconds
```

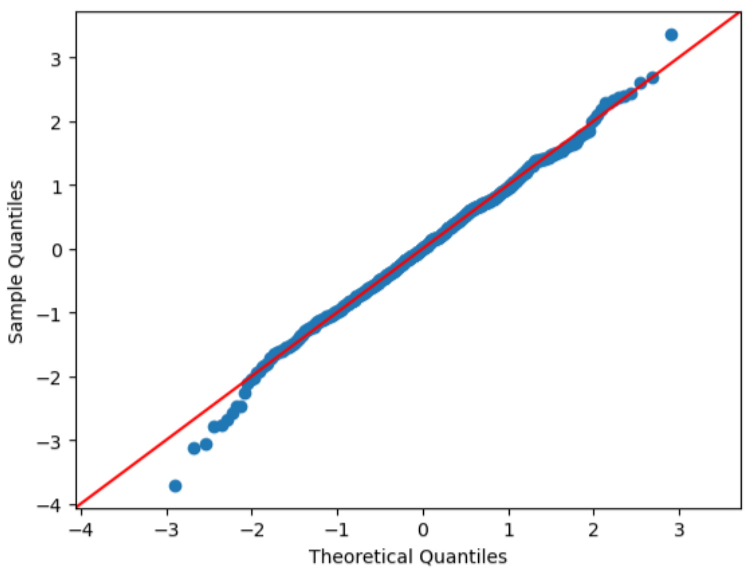
```
OLS Regression Results
==============================================================================
Dep. Variable:              log_price   R-squared:                       0.910
Model:                            OLS   Adj. R-squared:                  0.909
Method:                 Least Squares   F-statistic:                     896.2
Date:                Fri, 06 Mar 2026   Prob (F-statistic):         1.04e-274
Time:                        22:13:23   Log-Likelihood:                 216.98
No. Observations:                 540   AIC:                            -420.0
Df Residuals:                     533   BIC:                            -389.9
Df Model:                           6
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
const          6.3922      0.043    147.494      0.000       6.307       6.477
bedrooms       0.0363      0.011      3.307      0.001       0.015       0.058
area           0.0006   1.07e-05     55.182      0.000       0.001       0.001
stories        0.1407      0.009     15.045      0.000       0.122       0.159
bathrooms      0.1546      0.016      9.785      0.000       0.124       0.186
basement       0.1570      0.015     10.242      0.000       0.127       0.187
parking        0.0600      0.009      6.797      0.000       0.043       0.077
==============================================================================
Omnibus:                        4.610   Durbin-Watson:                   1.959
Prob(Omnibus):                  0.100   Jarque-Bera (JB):                5.052
Skew:                          -0.121   Prob(JB):                       0.0800
Kurtosis:                       3.407   Cond. No.                     2.23e+04
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly spec
ified.
[2] The condition number is large, 2.23e+04. This might indicate that there are
strong multicollinearity or other numerical problems.
```

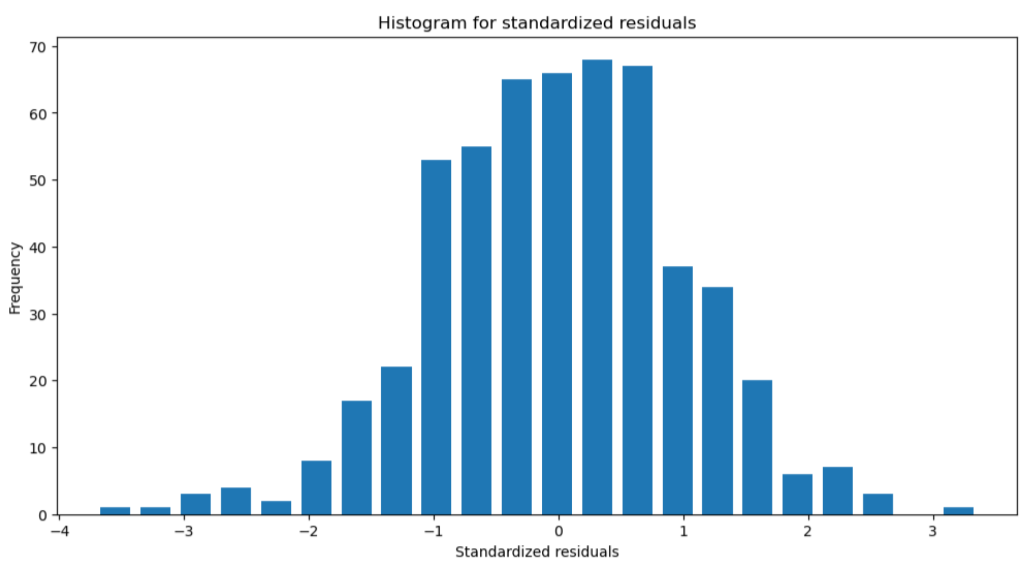
The significant variables after feature selection are bedrooms, area, stories, bathrooms, basement, and parking with p-value is around 0.000, which is less than 0.05 significance.

The regression equation:
Log_price = 6.3922 + 0.0363bedrooms + 0.0006area + 0.1407stories + 0.1546bathrooms + 0.1570basement + 0.0600parking

R-squared: 0.910
AIC = -420
BIC = -389.9


Residuals vs. Fitted values
Residuals
Fitted values


Sample Quantiles
Theoretical Quantiles


Histogram for standardized residuals
Frequency
Standardized residuals

The forward feature selection method has selected significant variables as mentioned above. However, the slight left skewness still appears and the diagnostic plots are mostly the same as those in step 7. This leads to the solution of checking potential outliers and excluding it.

<u>Step 6: Check potential and exclude them. Do the diagnostic checks for the new data.</u>

Using the Interquartile Range (IQR) to identify outliers because the data is skewed due to significant outliers. The following table is the list of outliers that are recorded in the data.

	area	bedrooms	bathrooms	stories	basement	parking	log_price
120	6244.997998	3	1	1	0	2	10.139901
145	3583.294573	3	1	2	1	0	8.756768
157	4062.019202	4	2	2	1	0	10.219759
406	2315.707235	3	1	3	0	0	7.936753
407	2554.407955	3	1	2	1	0	7.996794
469	3298.484500	4	1	2	0	1	8.478565

The following table is the data after removing outliers.

```
data4 = data3[(residual_norm3 > Q1-1.5*iqr) & (residual_norm3 < Q3+1.5*iqr)]
data4
```

	area	bedrooms	bathrooms	stories	basement	parking	log_price
0	4306.971093	4	2	3	0	2	10.077761
1	4732.863826	4	4	4	0	3	10.740163
2	4989.989980	3	2	2	1	2	10.146089
3	4330.127019	4	2	2	1	3	9.849127
4	4306.971093	4	1	2	1	2	9.774687
...	...	...	...	...	...	...	...
535	2738.612788	2	1	1	1	2	8.534901
536	2449.489743	3	1	1	0	0	8.044885
537	3008.321791	2	1	1	0	0	8.561375
538	2697.220792	3	1	1	0	0	8.448412
539	3102.418411	3	1	2	0	0	8.790602

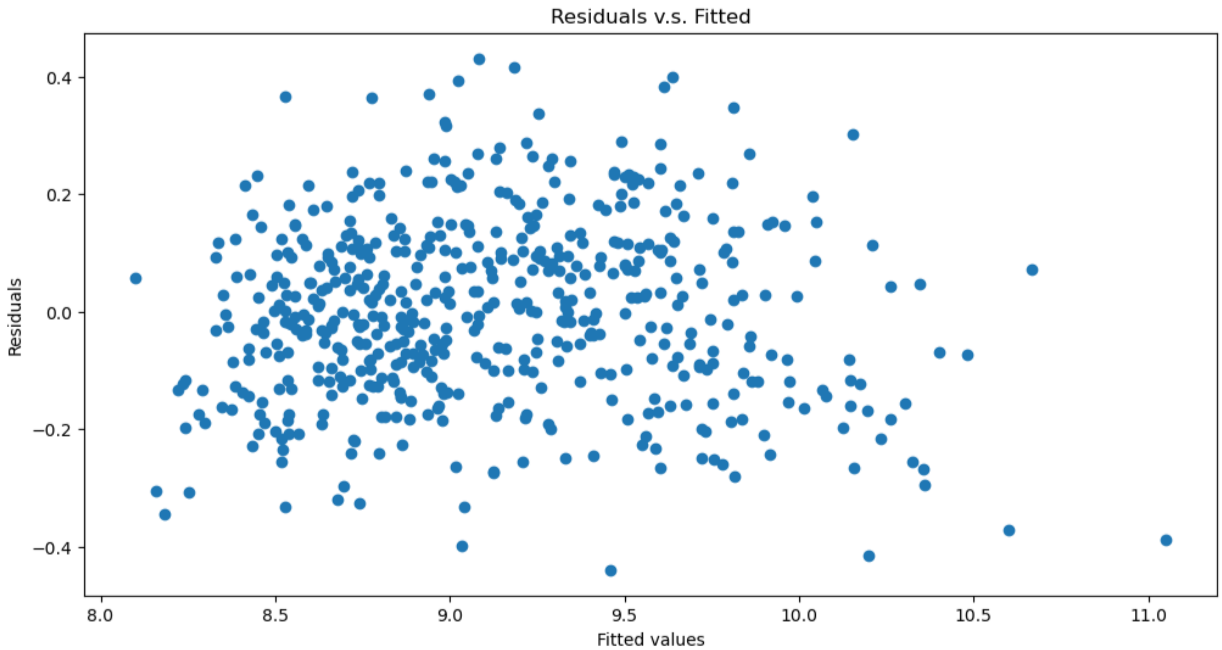
534 rows × 7 columns

Construct diagnostic checks

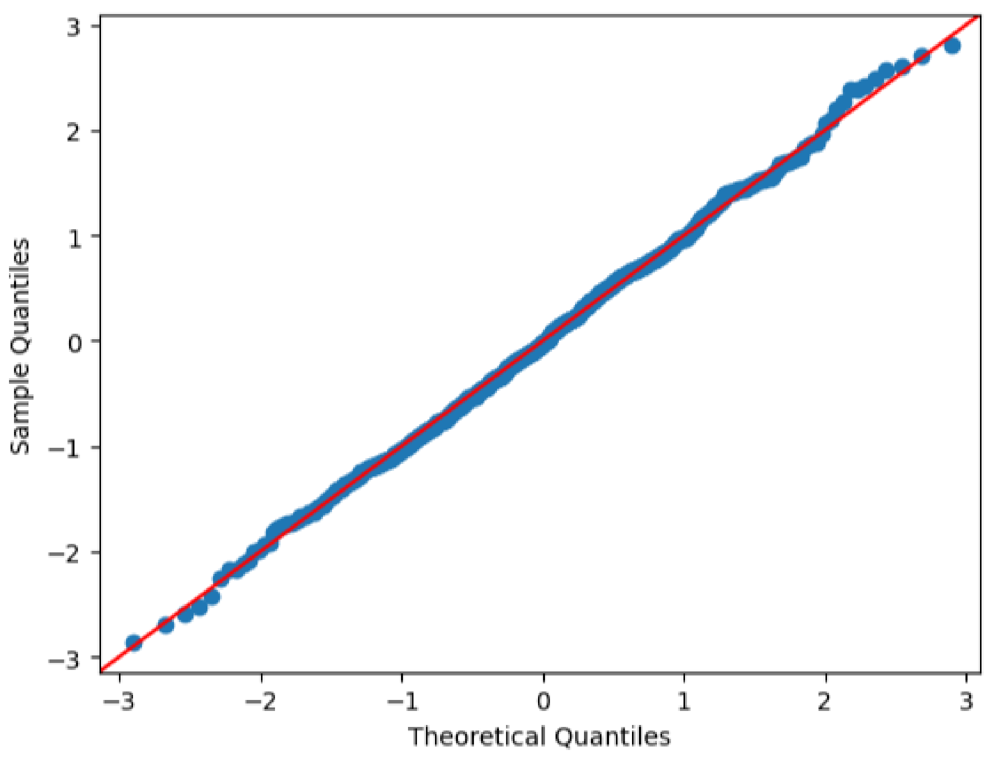
```
OLS Regression Results
==============================================================================
Dep. Variable:              log_price   R-squared:                       0.917
Model:                            OLS   Adj. R-squared:                  0.916
Method:                 Least Squares   F-statistic:                     970.4
Date:                Fri, 06 Mar 2026   Prob (F-statistic):          4.46e-281
Time:                        22:13:24   Log-Likelihood:                 243.18
No. Observations:                 534   AIC:                            -472.4
Df Residuals:                     527   BIC:                            -442.4
Df Model:                           6
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
const          6.4023      0.041    154.283      0.000       6.321       6.484
area           0.0006   1.03e-05     57.128      0.000       0.001       0.001
bedrooms       0.0377      0.010      3.613      0.000       0.017       0.058
bathrooms      0.1439      0.015      9.574      0.000       0.114       0.173
stories        0.1433      0.009     16.090      0.000       0.126       0.161
basement       0.1573      0.015     10.744      0.000       0.129       0.186
parking        0.0612      0.008      7.285      0.000       0.045       0.078
==============================================================================
Omnibus:                        0.799   Durbin-Watson:                   1.998
Prob(Omnibus):                  0.671   Jarque-Bera (JB):                0.886
Skew:                           0.043   Prob(JB):                        0.642
Kurtosis:                       2.820   Cond. No.                     2.24e+04
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly spec
ified.
[2] The condition number is large, 2.24e+04. This might indicate that there are
strong multicollinearity or other numerical problems.
```

The result indicates that excluding outliers slightly improves the data statistic information. R-squared increases from 0.910 to 0.917 and AIC, BIC are even smaller than the previous result. Moreover, the regression has the same significant variables as before, which are area, bedrooms, bathrooms, stories, basement, and parking.

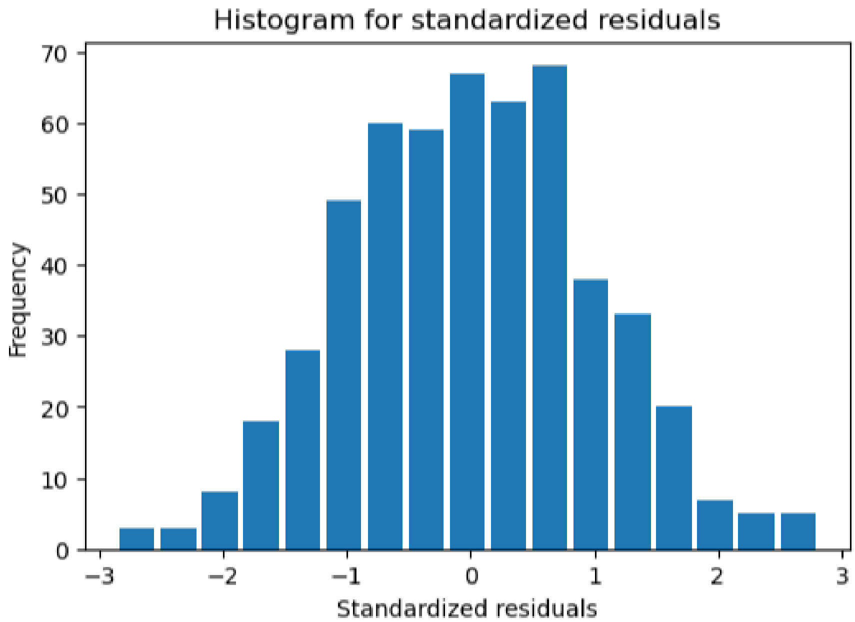
The regressions equation:
log_price = 6.4023 + 0.0006area + 0.0377bedrooms + 0.1439bathrooms + 0.1433stories + 0.1573basement + 0.0612parking



The residual plot shows no violation on either zero mean assumption or constant variance assumption as the step before.



The close alignment of data points to the red reference line validates the normality assumption, suggesting the current model is accurate with no significant skewness.



The residuals appear to be a normal distribution in the histogram plot, indicating the validation of the normality assumption and the accuracy of the model to use to predict test data.

Question 1. Do guestroom, basement, hot water heater, and air conditioning significantly influence housing price (significance), and how (magnitude)?

Only the basement significantly influences the housing price. Houses with a basement are expected to have prices approximately 17.03% higher than houses without a basement, holding other variables constant.

The guestroom, hot water heater and air conditioning are not significant factors that influence housing price.

<u>Step 7: Predict the test data using the current model</u>

This is the original test data:

	area	bedrooms	bathrooms	stories	mainroad	guestroom	basement	hotwaterheating	airconditioning	parking	price
0	3391.164992	3	2	2	yes	yes	no	no	yes	2	13648.38942
1	4006.245125	3	2	2	yes	no	no	no	yes	1	17305.14429
2	3286.335345	3	1	2	no	no	yes	yes	no	2	10205.51922
3	4229.361654	3	2	1	yes	yes	yes	no	yes	2	17321.41596
4	4486.089611	3	1	1	yes	yes	yes	no	yes	1	15862.92477

The test data after convert string variables into dummy variables:

	area	bedrooms	bathrooms	stories	mainroad	guestroom	basement	hotwaterheating	airconditioning	parking	price
0	3391.164992	3	2	2	1	1	0	0	1	2	13648.38942
1	4006.245125	3	2	2	1	0	0	0	1	1	17305.14429
2	3286.335345	3	1	2	0	0	1	1	0	2	10205.51922
3	4229.361654	3	2	1	1	1	1	0	1	2	17321.41596
4	4486.089611	3	1	1	1	1	1	0	1	1	15862.92477

Question 2. Make prediction on the test set *csv*, and report the mean square error (MSE).

```
from sklearn.metrics import mean_squared_error

x_test = test1.iloc[:, [0,1,2,3,6,9]]
x_test = sm.add_constant(x_test, has_constant = "add")

y_test = test1.iloc[:,10]

y_pred = np.exp(lm4.predict(x_test))

mse = mean_squared_error(y_test, y_pred)
print(f"The mean squared error is: {mse}")
```

```
The mean squared error is: 5970016.947270194
```

The mean squared error is 5970016.9473, which corresponds to and RMSE $2,444. This is a large deviation to the test housing price, indicating the model's prediction deviate substantially from the actual values. However, the test set only contains 4 observations, this high MSE does not reflect fully the model's overall predictive performance. Since all the assumptions were checked, the model demonstrates a proper modeling procedure in order to get the most accurate model.